

What I added to the dataset, and why

42 columns to 111. Every rule, a worked example, and an honest source

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My dataset started with 42 columns. It now has **111**. This document explains every group of new columns: what the word means, one real example from my own data, and **where the rule came from**, with a link.

I also grade every rule honestly, because my supervisor will ask “are you sure about this?” and the truthful answer is different for different columns.

1. First, the one rule that makes all of this possible

Every one of these columns is read out of the **ingredient list** printed on the back of the box. In the industry that list is called the **INCI list**.

Two facts about that list do the heavy lifting.

Fact one: the order is not random. EU law says ingredients must be listed from most to least. Below 1% the order stops mattering, so anything after the 1% mark can be in any order.

“The list of ingredients shall be established in descending order of weight of the ingredients at the time they are added to the cosmetic product. Ingredients in concentrations of less than 1 % may be listed in any order.”

That is [Regulation \(EC\) No 1223/2009, Article 19\(1\)\(g\)](#). This is **law**, not an opinion.

Why I care: if I see RETINOL at position 4 out of 8, it is a real dose. If I see it at position 34 out of 40, it is nearly nothing. That is where my `lead_active_position` column comes from.

Fact two: nano ingredients must be labelled. The same article says nanomaterials must be followed by the word “(nano)” in brackets. So I can detect them by looking for that exact bracket.

What this means for the thesis: I am not guessing. I am reading a legally-mandated, legally-ordered document. My supervisor can verify any single cell by picking up the physical product.

I did no new scraping for any of this. No API calls, nothing bought. It is all squeezed out of data I already had.

2. How to read the grade next to every column

This is the honest part. Not all 69 columns rest on the same quality of evidence. So every group below carries one of four grades.

Grade	What it means	Can my supervisor challenge it?
A — Law	An EU regulation with an article number	No. It is legislation
B — Published science	A named paper in a named journal	Only on interpretation
C — Industry practice	What the cosmetic analysis sites do. No single paper behind it	Yes, and fairly
D — My own rule	I invented it for convenience	Yes. It is a label, not a finding

If you only remember one thing: grades A and B are defensible in a viva. Grade C is common practice that I should describe as common practice. Grade D I should call a convenience label and not defend as science.

Rough split: **A covers 9 columns, B covers 21, C covers 26, D covers 13.**

3. The groups, one by one

Group 1. Allergy and dietary avoidance

Grade A for the allergen link, Grade C for the rest

Columns: contains_animal_derived, contains_tree_nut, contains_gluten_grain, contains_soy, contains_coconut, plus an _evidence column for each.

What it means: the product contains something from an animal, or something a person with a nut, gluten or soy issue may want to avoid.

Real example. A product with CERA ALBA in the list gets contains_animal_derived = yes and the evidence cell says CERA ALBA. Cera alba is the Latin INCI name for beeswax.

Where it comes from. The names themselves are legally controlled. The EU publishes **CosIng**, the official ingredient database, and my dataset is already matched to it for 99.2% of products with an ingredient list. Latin plant and animal names in INCI are not free text.

Numbers: animal 1,641 (13.0%), tree nut 1,243 (9.8%), gluten grain 861 (6.8%), soy 815 (6.5%), coconut 2,778 (22.0%).

Related and stronger: the fragrance allergen list. This is Grade A and it is a real opportunity for my thesis. EU law used to require 26 named fragrance allergens to be declared. That changed. [Commission Regulation \(EU\) 2023/1545](#) expanded the list from 26 to **over 80 substances**, adding 56 new ones. The deadlines are **31 July 2026** and **31 July 2028**.

Say this to my supervisor: the allergen rules are changing right now, in the same year I am submitting. No skincare ontology in my literature review has the new list. That is a live gap, not a historical one.

Group 2. Malassezia, also called fungal acne

Grade B, but with an honest warning

Columns: feeds_malassezia, malassezia_evidence.

What the word means. Malassezia is a **yeast that lives on everybody's skin**. It is normal. In some people it overgrows and causes small itchy bumps that look like acne but are not acne. Online people call it "fungal acne". The medical names are pityrosporum folliculitis and seborrhoeic dermatitis.

Why an ingredient can make it worse. This is the good part, and it is real biology. Malassezia **cannot make its own fatty acids**. It is missing the genes for it. So it has to eat fatty acids from its surroundings, and it secretes lipases to break down oils and get them.

Which fatty acids? Ones roughly **11 to 24 carbon atoms long**, with the best documented feeding around C16 to C18.

Source: ["Lipid-dependent growth of Malassezia spp. in defined medium with single fatty acids", FEMS Yeast Research, 2025](#). Open access copy on [PubMed Central](#).

Real example. The Ordinary Retinol 0.2% in Squalane gets feeds_malassezia = yes, evidence CAPRYLIC/CAPRIC TRIGLYCERIDE. That ingredient is a C8 to C10 oil, so honestly this one is borderline and sits at the edge of the range.

My honest warning. 7,093 products, **56.2%**, come out as yes. That is more than half the catalogue, which tells you the flag is broad. The underlying biology is solid, but the exact list of which cosmetic ingredients matter in a real formula, at real concentrations, is **not settled science**. SkinCarisma, the site most people use for this, says so itself.

What I will say in the viva: the mechanism is published and I cite it. The ingredient list is a screening heuristic. It should be shown to a user as "contains ingredients some people avoid", not as "this will break you out".

Group 3. Comedogenic, meaning pore-clogging

Grade B for the concept, Grade C for the list, and the science is contested

Columns: contains_comedogenic, comedogenic_evidence.

What the word means. Comedogenic means an ingredient tends to block a pore and form a comedone, which is the blackhead or whitehead that starts acne.

Where this whole idea came from, and this is a good story for the viva.

In **1972** two dermatologists, **Albert Kligman and Otto Mills**, published a paper called *Acne Cosmetica*. They noticed adult women getting closed comedones and blamed their cosmetics. To test ingredients, researchers then developed the **rabbit ear assay**: paint the substance inside the ear of an albino rabbit for two to four weeks, then look at the skin under a microscope and count the comedones. The big published ingredient tables everyone still quotes come from this. See [Kligman's 1984 ratings of therapeutic products, cosmetics and ingredients in the rabbit ear, JAAD](#).

Here is the problem, and I should say it out loud. In 1982 Kligman himself tested the same chemicals **on humans** and found that some strong comedogens in rabbits produced little or no comedones in people. The rabbit model is **oversensitive**. There is a full re-evaluation of the concept in [JAAD, 2005](#), and a [2025 clinical review in JAAD Reviews](#) specifically about the regulatory gaps.

Number: 1,991 products, 15.8%.

Say this to my supervisor: "the comedogenic ratings everybody uses come from rabbit ears in the 1970s and 80s, the man who invented the method showed himself in 1982 that it overpredicts in humans, and I am labelling my column accordingly." That sentence alone shows I read past the surface.

Group 4. Barrier repair

Grade B, and this is my strongest scientific column

Columns: supports_barrier_repair, barrier_repair_evidence, contains_ceramide.

What "barrier" means. The outermost layer of skin, the stratum corneum, is often described as a brick wall. The dead skin cells are the bricks, and a mortar of fats holds them together. That mortar keeps water in and irritants out. When it is damaged, skin stings, flakes and reacts to everything.

What the mortar is made of. Three lipids: **ceramides, cholesterol and free fatty acids**.

The published finding. Work by **Peter Elias and colleagues at UCSF** showed that if you apply only one or two of these lipids to damaged skin, you **delay** recovery. Only a complete mixture allows normal repair. And repair gets faster when one of the three is increased up to threefold, which is where the famous **3:1:1 ratio** comes from.

Source: Man, Feingold, Thornfeldt and Elias, "Optimization of Physiological Lipid Mixtures for Barrier Repair", *Journal of Investigative Dermatology*, and the follow-up on aged skin in *JAAD*.

Real example. CeraVe Moisturising Face Lotion SPF30 gets supports_barrier_repair = yes, evidence NIACINAMIDE.

Numbers: barrier repair 5,690 (45.1%), ceramide specifically 1,064 (8.4%).

Where I should be careful. My marker list is wider than the three lipids. I also count niacinamide, panthenol, centella and squalane, which are supported but are soothing agents rather than barrier lipids. **A stricter version of this column that only counts the three Elias lipids would be more precise**, and it is an easy improvement if a supervisor pushes.

Group 5. Retinoids, night use and layering conflicts

Grade B

Columns: contains_retinoid, retinoid_evidence, use_time, do_not_layer_with, may_cause_purging, purging_evidence.

What a retinoid is. A vitamin A derivative. It speeds up how fast skin cells turn over. It is the single best-evidenced anti-ageing ingredient there is, and also the most irritating.

Why night only. Retinoids break down in light. When retinoid compounds are hit by UV they isomerise and oxidise, which destroys the molecule and its function. One study found isotretinoin drops significantly after roughly one hour's worth of sunlight exposure.

Source: [Photostability of Topical Agents Applied to the Skin: A Review, Pharmaceutics 12\(1\):10, MDPI, 2020.](#)

What "purging" means. When you start a retinoid or an acid, skin can look **worse for four to eight weeks** before it gets better, because everything already forming under the surface is being pushed up faster. It is not an allergic reaction and it is not the product failing. Users quit at week three because nobody told them. That is exactly the sort of thing a recommender should say.

Real example. The Ordinary Retinol 0.2% in Squalane: use_time = night only, may_cause_purging = yes with evidence RETINOL, and do_not_layer_with = acids (AHA, BHA) and vitamin C, on the same night.

Numbers: retinoid 681 (5.4%), purging 2,510 (19.9%), layering warning 3,002 (23.8%).

Group 6. Exfoliating acids

Grade C, with one important correction

Columns: contains_aha, aha_evidence, contains_bha, bha_evidence, contains_pha.

The terms.

Term	Full name	In one line
AHA	alpha hydroxy acid	Water-loving. Works on the surface. Glycolic, lactic
BHA	beta hydroxy acid	Oil-loving, so it gets down inside a pore. Salicylic
PHA	poly hydroxy acid	Bigger molecule, goes in slower, gentler. Gluconolactone

The correction, and this one matters. My first version counted **citric acid** as an AHA. Technically it is one. But in practice it is in almost every formula as a **pH adjuster**, at a level that does nothing to exfoliate. Counting it would have wrongly labelled **2,842 products** as exfoliants. I took it out. The AHA count dropped from 4,096 to **1,254**.

Numbers: AHA 1,254 (9.9%), BHA 1,026 (8.1%), PHA 409 (3.2%).

Group 7. Mature skin

Grade C, and I think this column is too generous

Columns: suits_mature_skin, mature_skin_evidence.

What "mature skin" actually means, because a friend asked me this and I did not have a clean answer.

It is a **trade term, not a medical diagnosis**. There is no age at which it switches on. It describes skin that has lost collagen and lipids over time, so:

- fine lines and loss of firmness, because collagen production slows
- **drier**, because oil production drops after menopause
- slower cell turnover, so it looks duller and heals slower
- more uneven pigment from accumulated sun exposure
- thinner and more fragile

The practical difference: a product for oily acne-prone skin and a product for mature skin can target the same concern and need completely opposite bases. Mature skin usually needs the richer base.

Real example. CeraVe Moisturising Face Lotion gets `suits_mature_skin = yes` on evidence NIACINAMIDE.

My honest problem with this column. 4,802 products, **38.0%**, say yes. That is too many to be useful as a filter, and the reason is that I included **niacinamide**, which is in a huge number of products. A stricter version counting only retinoids, peptides and vitamin C would give a smaller and much more meaningful number. **I would flag this myself before a supervisor finds it.**

Group 8. EU regulatory watch

Grade A. This is the strongest group in the whole dataset

Columns: `contains_nanomaterial`, `contains_solid_microplastic`, `contains_synthetic_polymer`, `contains_endocrine_concern`, `contains_pfas`, plus evidence columns.

Why this group is special. These are not opinions or industry habits. These are **named EU regulations**, several of which land in 2026 and 2027, which is exactly when I submit. **Not one paper in my literature review has this.**

Column	The law	What it does
<code>contains_nanomaterial</code>	1223/2009 Art. 19(1)(g)	Nano ingredients must be labelled "(nano)". I detect the bracket
<code>contains_solid_microplastic</code>	Regulation (EU) 2023/2055	Restricts solid synthetic polymer microparticles
<code>contains_endocrine_concern</code>	Ongoing SCCS review and restriction	Substances under active EU review as endocrine disruptors
<code>contains_pfas</code>	Proposed EU-wide restriction	The "forever chemicals"

The correction I had to make. My first version counted **carbomer** as a microplastic and got 3,125 hits, of which 1,930 were carbomer. That is wrong. Carbomer is a **swollen gel, not a solid bead**, and 2023/2055 restricts solid particles. So I split the column in two:

- `contains_solid_microplastic` = **589** (4.7%). Actually in scope of the law
- `contains_synthetic_polymer` = **3,166** (25.1%). Broader, informational only

Numbers: nano 101 (0.8%), solid microplastic 589 (4.7%), synthetic polymer 3,166 (25.1%), endocrine concern 1,582 (12.5%), PFAS 51 (0.4%).

Group 9. Sunscreen chemistry

Grade C for the classification, Grade B for the reef flag

Columns: `sunscreen_filter_type`, `contains_reef_harmful_uv`, `reef_harmful_evidence`, `white_cast_risk`.

The terms. A **mineral** filter, zinc oxide or titanium dioxide, sits on top and reflects. A **chemical** filter, like avobenzone, absorbs UV and turns it into heat. **Hybrid** means both.

White cast is the grey film mineral sunscreens can leave. It shows more on deeper skin tones. It is the single biggest reason people stop wearing sunscreen, so it belongs in a recommender.

Reef harmful flags oxybenzone and octinoxate, which are banned in Hawaii, Palau and Key West.

Numbers: filter type on 1,883 products (14.9%), reef harmful 747 (5.9%), white cast risk 1,056 (8.4%).

Group 10. Formula shape

Grade C

Columns: formula_base, moisturiser_type, preservative_system, texture, contains_fragrance, contains_essential_oil, contains_drying_alcohol.

The terms that matter.

Water-based versus anhydrous. Anhydrous means no water. This is not trivia: **anything with water needs a preservative**, and anything without water usually does not. It is also why an anhydrous oil and a water serum behave differently when layered.

Moisturiser type. A **humectant** draws water in, like glycerin or hyaluronic acid. An **occlusive** seals water in, like petrolatum. “Draws and seals” means the formula does both, which is what a good moisturiser should do. In dry winter air a humectant with nothing sealing it can actually pull water **out** of the skin.

Drying alcohol, and a correction. My first version counted **cetearyl alcohol** as a drying alcohol. That is wrong. Cetearyl, cetyl and stearyl are **fatty alcohols**, which are waxy and conditioning, the opposite of drying. Only alcohol denat, SD alcohol and ethanol count. The number fell from 1,503 to **826**.

Real example. CeraVe: formula_base = water-based, moisturiser_type = draws and seals, preservative_system = Organic acid, Glycol or alcohol, texture = lotion. The Ordinary retinol: formula_base = anhydrous, preservative_system = none detected, which is correct and expected for an oil.

Numbers: fragrance 5,047 (40.0%), essential oil 1,888 (14.9%), drying alcohol 826 (6.5%), texture 7,432 (58.8%).

Group 11. Routine placement

Grade D. My own rules

Columns: routine_step, body_site, use_time, strength_level, regional_style, price_band_lebanon.

What they do. routine_step puts each product in order, so “1 Cleanse” through “7 Protect”. strength_level says beginner, intermediate or advanced. regional_style marks Korean, French pharmacy, Lebanese local and so on.

I am being straight here: I invented these. There is no standard behind them. They are convenience labels that make the recommender usable. They are **not findings** and I will not defend them as science. If asked, the correct answer is “these are presentation rules I defined, and they are documented in the script”.

routine_step and body_site are on 100% of products. price_band_lebanon is on 92.6%, and that one is grounded in my own real price data, so it is stronger than the rest of this group.

Group 12. Brand ownership

Grade B. Verifiable public fact

Column: brand_parent_company.

Kenvue owns Neutrogena. L'Oreal owns CeraVe. Unilever owns Paula's Choice. These are checkable against company annual reports.

Why it is useful: it shows how concentrated my catalogue really is, and it lets any brand-level question be answered later without touching the dataset.

Coverage is only 3,109 products, 24.6%. This is the weakest coverage of any column I kept. Wikidata has a parent organization property that would raise it a lot in one pass. That is already on my Sprint 2 list.

(Note: I built and then removed a boycott-listing feature. The published BDS priority list matched only 2 of my 12,629 products, so it was not worth a column. The parent company data survived because it is useful on its own.)

4. Two full worked examples

Example A. CeraVe Moisturising Face Lotion SPF30 AM, 52ml

40 ingredients. Here is what the system now knows without anyone typing anything:

Column	Value	Read it as
formula_base	water-based	Has water, so needs preservative
preservative_system	Organic acid, Glycol or alcohol	And it has one
moisturiser_type	draws and seals	Humectant plus occlusive. A proper moisturiser
contains_ceramide	yes	The brand's whole selling point, confirmed from the actual list
supports_barrier_repair	yes, via NIACINAMIDE	Good for damaged, stinging skin
suits_mature_skin	yes, via NIACINAMIDE	But see my warning in Group 7
feeds_malassezia	yes, via ISOPROPYL PALMITATE	Fungal acne sufferers should skip it
contains_comedogenic	yes, via ISOPROPYL PALMITATE	Same ingredient, two different flags
contains_fragrance	no	Safe for the fragrance-sensitive
routine_step	6 Moisturise	Where it goes
use_time	morning	It has SPF
strength_level	intermediate	
actives_present	niacinamide	
lead_active_position	niacinamide at position 6	Position 6 of 40, so a real dose, not a sprinkle

Notice the useful tension. This is a barrier-repair product that is also flagged for fungal acne, and both flags point at the same single ingredient, isopropyl palmitate. A keyword search would never surface that. A rule over a structured ingredient list does.

Example B. The Ordinary Retinol 0.2% in Squalane

Only 8 ingredients.

Column	Value	Read it as
formula_base	anhydrous	No water
preservative_system	none detected	Correct. No water means no preservative needed
contains_retinoid	yes, RETINOL	
use_time	night only	Retinol degrades in UV. Source in Group 5
may_cause_purging	yes, RETINOL	Warn the user it may get worse for 4 to 8 weeks
do_not_layer_with	acids (AHA, BHA) and vitamin C, same night	The conflict warning
strength_level	advanced, patch test first	
suits_mature_skin	yes, RETINOL	This time on strong evidence, unlike Example A
lead_active_position	retinol at position 4	Position 4 of 8. Genuinely concentrated
supports_barrier_repair	yes, SQUALANE	The squalane in the name

Why this example is worth showing. Four of these columns are safety information a shopper actually needs and will not find on the box: do not use in daylight, do not combine with acids tonight, expect it to look worse first, patch test. That is what an ontology is for.

5. Three things I could not do, and why

My supervisor suggested pulling organic, cruelty free and vegan from what I already had. I tried, and I want to be clear about why it failed, because the failure is itself worth writing up.

First I tried mining the text. My product_summary column looked promising. It is not. It turns out to be **self-generated boilerplate**, the same phrases repeated. Searching it for “cruelty free” returned **16 products out of 12,629**. That is noise, not a signal.

Then I understood why it can never work from ingredients.

Claim	Why the INCI list cannot tell you
Cruelty free	It is about whether the finished product or its ingredients were tested on animals . That is a fact about a company’s process, not about a molecule. Two identical formulas can differ
Organic	It is about how the plant was farmed . Organic and conventional lavender oil have the exact same INCI name
Vegan	Closer, since I can detect animal-derived ingredients. But absence of evidence is not evidence of absence. I can prove a product is not vegan. I cannot prove it is

The rule I followed everywhere: positive detection only. I say what IS present. I never claim something is absent, because an ingredient list I failed to parse looks identical to a clean formula.

This is the right answer, not a cop-out. All three of these are **certifications by an external body**, which is a completely different kind of information from an ingredient. The correct way to model them is as a claim by a named certifier with a date, not as a property of the product.

6. Five mistakes I found and fixed

I want these in the document because “I tested it and found my own errors” is a stronger thing to say than “it worked first time”.

#	The mistake	How I found it	The fix
1	Synthetic beeswax counted as animal-derived, 43 products	Read random matches by eye	Added an exclusion list: synthetic, vegetal, biomimetic, vegan
2	Cetearyl alcohol counted as drying alcohol	Same	Excluded fatty alcohols. 1,503 down to 826
3	Citric acid counted as exfoliating AHA	Count looked far too high	Removed it. Would have mislabelled 2,842 products . 4,096 down to 1,254
4	Carbomer counted as a solid microplastic, 1,930 of 3,125 hits	Read the regulation properly	Split into two columns. 589 in scope, 3,166 informational
5	The script silently died. It recompiled the same patterns for every ingredient of every product, roughly 40 million times	It just never finished	Cached the compiled patterns. Minutes down to 8 seconds

The method that caught all five: after every rule, pull random matching products and read them. Four of the five would have survived any amount of code review, because the code was correct. The **rule** was wrong.

7. Short dictionary

Term	Plain meaning
INCI	International Nomenclature of Cosmetic Ingredients. The standard naming system. Why every box says “Aqua” not “water”
CosIng	The European Commission’s official cosmetic ingredient database
Anhydrous	Contains no water
Humectant	Draws water into skin. Glycerin, hyaluronic acid
Occlusive	Seals water in. Petrolatum, dimethicone
Comedone	A blocked pore. A blackhead or whitehead
Comedogenic	Tends to block pores
Malassezia	A yeast that lives on normal skin. Can overgrow into itchy bumps
Stratum corneum	The outermost layer of skin. The “barrier”
Ceramide	One of the three fats that hold the barrier together
Retinoid	A vitamin A derivative. Speeds cell turnover
Purging	Skin temporarily worsening when starting a retinoid or acid. 4 to 8 weeks
AHA / BHA / PHA	Three families of exfoliating acid. Water-loving, oil-loving, gentle
Mature skin	Trade term for skin that has lost collagen and lipids. Not an age
Fatty alcohol	Waxy and conditioning. The opposite of drying alcohol despite the name

Term	Plain meaning
Nanomaterial	Particles small enough that EU law requires “(nano)” on the label
Fragrance allergen	One of the substances EU law requires to be named separately. Was 26, now over 80
SCCS	Scientific Committee on Consumer Safety. The EU’s expert advisory body

8. All sources in one list

Law, Grade A

1. [Regulation \(EC\) No 1223/2009, Article 19](#). Ingredient order, the 1% rule, nano labelling
2. [Commission Regulation \(EU\) 2023/1545](#). Fragrance allergens from 26 to over 80. Deadlines 31 July 2026 and 31 July 2028
3. [Regulation \(EU\) 2023/2055](#). Solid synthetic polymer microparticles
4. CosIng, the European Commission ingredient database. Already linked for 99.2% of my products with an ingredient list

Published science, Grade B

5. [Lipid-dependent growth of *Malassezia* spp. in defined medium with single fatty acids](#), FEMS Yeast Research, 2025 (PMC copy)
6. Kligman and Mills, *Acne Cosmetica*, 1972. The origin of the whole comedogenic idea
7. [Comedogenicity of current therapeutic products, cosmetics, and ingredients in the rabbit ear](#), JAAD, 1984
8. [A re-evaluation of the comedogenicity concept](#), JAAD, 2005
9. [Comedogenicity in cosmeceuticals: clinical relevance, regulatory gaps and future directions](#), JAAD Reviews, 2025
10. [Man, Feingold, Thornfeldt and Elias, Optimization of Physiological Lipid Mixtures for Barrier Repair](#), Journal of Investigative Dermatology
11. [Optimal ratios of topical stratum corneum lipids improve barrier recovery in chronologically aged skin](#), JAAD
12. [Photostability of Topical Agents Applied to the Skin: A Review](#), Pharmaceutics 12(1):10, 2020

Industry practice, Grade C. Named honestly as practice, not as science.

13. SkinCarisma. *Malassezia*, comedogenic, silicone, alcohol and allergen flags
14. CosDNA. Comedogenic and irritancy ratings
15. SkinSort and EWG. The “free from” vocabulary consumers expect

Grade D. My own rules: `routine_step`, `body_site`, `strength_level`, `regional_style`, `texture` and the presentation labels. Documented in `FINAL_PIPELINE/add_derived_attributes.py`. Convenience labels, not findings.

The one paragraph version

I did not invent a vocabulary. I took the properties that the cosmetic analysis sites already flag, added the EU regulations that land in 2026 and 2027, and derived all of them from the legally ordered ingredient list I already had. Where a real published paper backs a rule, I cite it. Where only industry practice backs it, I say so. Where I made the rule up, I say that too. Every flag carries an evidence cell naming the exact ingredient that triggered it, so any single value can be checked by hand in a few seconds.

42 columns to 111. 12,629 rows unchanged. Zero original cells altered.